

Sustainable Vertical Farming: Leveraging Machine Learning and IoT for Energy Efficiency and Productivity

Shruti Gupta
Computer Engineering
Poornima Institute of Engineering
and Technology
Jaipur, Rajasthan, India
shruti.gupta@poornima.org

Arnab Das
Computer Science Engineering
National Institute of Technology
Itanagar, Arunachal Pradesh, India
arnab.phd24@nitap.ac.in

Sanjay Kumar Sinha
Computer Engineering
Poornima Institute of Engineering &
Technology
Jaipur, Rajasthan, India
sanjay.sinha@poornima.org

Abstract

Vertical farming as an idea has emerged as a helpful strategy to mitigate problems related to food production throughout the world, less land available, and climate changes. Through the use of controlled environment agriculture, vertical farming consists of growing food in layers and within the vicinity of consumers, hence decreasing the need for conventional agriculture. One of the precision farming efficient crop growth practices that easily integrates machine learning (ML) technologies, and the Internet of Things (IoT) is vertical farming. Various technologies are used, including robotics, machine learning, big data analytics, drones, and IoT sensors in gathering data on crop growth, environmental conditions, and weather patterns. For this reason, energy efficiency is regarded as a significant parameter that determines the economic sustainability of vertical farms if not their environmental impacts. This article explores energy efficiency within vertical farming systems highlighting how technology, better practices, and the use of clean energy sources help counter the high energy needs of vertical farms. The paper also looks into energy-efficient systems and designs such as LED lights which consume lower electric power and are designed to produce light like natural sunlight, energy-efficient HVAC systems which operate with less energy, systems for recycling water and nutrients and minimize usage of these resources.

Keywords

Vertical farming, sustainability, energy efficiency, water management, machine learning

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1 Introduction

A tremendous rise in population is going to cause huge variations that the world will go through ecological issues in the next fifty

years. The global population was at 7.6 billion and is predicted to increase to 8.6 billion by 2030 and 9.8 billion by 2050 and 11.2 billion in 2100, according to a report published by the United Nations [1]. This increase in global population presents several risks, including increased food demands globally and more arable land. With growing populations globally and decreasing arable land, the demand for modern agricultural technologies is on the rise. This has led to the development of a technology called Vertical Farming—the art of growing crops in a pattern of layers or inclined surfaces. Figure 1, shows vertical farming in agriculture. Vertical farming reduces land use for the maximization of space and for utilizing crops all seasons without considering the seasons that may reduce the application of land, pesticides, and herbicides [2]. However, the general sustainability of this agricultural method greatly depends on the energy used by this process. Energy forms the core of the operation of vertical farms, literally from light, temperature, water circulation and even the nutrient delivery system. The agriculture as on a vertical farm is astonishing, holding great promise in producing healthier and more related nutrition with less transported food waste. The only drawback is the high energy usage. Artificial light, climate-friendly practices, and mechanization in average mean that losses in energy outweigh some benefits of ecological management if high control is not exercised [3].

The challenge of finding a balance that appropriately balances energy efficiency and productivity becomes critical when deciding to apply vertical farming in food production. More specifically, technological progress such as improved efficiency and effectiveness in LED growing lights, development of energy-efficient heating and cooling systems, and uses of alternate sources of energy have begun to significantly alter the energy profile of vertical farms [4]. However, there is now an imperative need for more innovations coupled with integrated approaches in order to heighten the production of vertical farming whilst lowering the carbon emissions associated with it.

This introduction will try to give it in an aspect that looks into the relation of energy use with vertical farming in reasonable extents of use but sustainable food production methods that would be worth the amount of additional consideration into the way in which this industry will manage to maintain and sustain its environmental concern in crop production over that hitherto farmed. It can be understood pretty well that vertical farming has immense development prospects within cities, where energy-efficient technologies will enable healthy, organic, local food supply with minimal environmental damage [5]. The temperature for plants will vary depending on the type of crop to be grown by vertical farming



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Figure 1: Agriculture in vertical farming

Table 1: Specification of Temperature ranges, LED Lighting Use and Energy Savings for different crop varieties

Plant Type	Optimal Temperature (°C)	LED Lighting Use	Energy Savings
Leafy Greens	15°C - 22°C	Tailored light spectrum for vegetative growth	40-70% reduction compared to HPS systems
Herbs	18°C - 24°C	Focus on blue/red light ratios for efficiency	Reduced heat output, lower cooling needs
Strawberries	20°C - 26°C	Enhanced full-spectrum LED lights	Reduced energy use in temperature control
Tomatoes	20°C - 26°C	Full-spectrum LEDs for optimal fruiting	Lower operational costs in lighting & HVAC
Flowers (Ornamental)	18°C - 22°C	Light designed for color balance & bloom growth	Better light control for aesthetic purposes

systems shown in Table 1. Crops must remain constant from the germination stage right up to the final yield stage of crops, where the temperature remains constant for each variety.

Traditional farming puts farmers in the most demanding situations, especially with respect to long working hours, disasters, unpredictable yields, and susceptibility of crops to pests and diseases [6]. The practice of vertical farming takes place in an entirely automated system that requires no human involvement. This is one of the latest developments in agriculture. This technology has brought positive results for the last few years. Numerous studies have been conducted to identify some specific applications to which vertical farming could be applied. The performance of vertical farming

systems can be enhanced with the help of IoT and ML technologies [7]. Based on the study, the paper is well-structured to understand the technologies related to vertical farming, such as literature study, design, types, comparative study, and performance analysis.

2 Literature Review

Technology in the last couple of years, which comprises of vertical farming, whereby there was an effort to raise crops akin if not of a much better quality, but was still dependent on conventional mechanisms to get it produced involves a variety of ways for instance being hydroponic, aquaponic, and aeroponic [8]. What distinguishes the root system from each one in that either it will

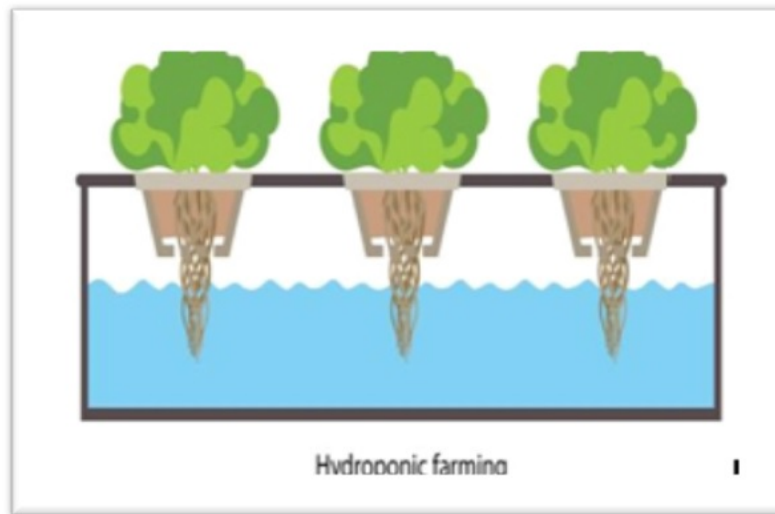


Figure 2: Hydroponic System

cover in fish habitat or liquids in nutrient solution-rich waters, or even expose them while under nourishing mist.

2.1 Hydroponics

In comparison to conventional cultivation, hydroponics can produce a higher yield by utilizing both the horizontal and vertical surfaces. Thus, to meet the daily demands of consumers in densely populated areas for nutritious, fresh products, the number of plants per unit area is being increased, and a shift towards hydroponic vertical farming is being adopted. Hydroponics also enables the annual production of a range of crops without the wanton release of either pesticides or fertilizers into the environment. Compared to traditional open-field farming, hydroponics requires less land and water. The hydroponic systems use a number of nutrient solution delivery techniques, including the deep water culture (DWC) ebb flow system, wick system, trickle, and nutrient film technique (NFT). The Deep Water Culture technique is a widely used hydroponic procedure in which the plant roots are immersed in an aerated nutrient solution. A hydroponic system that employs a water pump to periodically saturate the roots with the nutrient solution is known as the ebb and flow system [9]. Figure 2 shows Hydroponic system. This system is a flood and drain system. A water pump provides a nutrient solution to plants, thus allowing the roots to absorb moisture and assimilate nutrients. The roots can absorb oxygen during this period of excess nutrient solution emptying into a reservoir.

Whereas there are certain limitations for hydroponics, such as operating and maintaining it through the support of a very skillful manpower, water as well as air pump requirements tend to cause power failure occasionally. Moreover, high costs during establishment as well as in using more energy are included during this technique [10]. Hydroponics is also likely to yield greater greenhouse emissions since the method consumes more energy. However, this can be prevented by using renewable sources of energy and longer-lasting materials. The use of much machinery, irrigation

systems, and transportation consumes much energy. Such activities also have a huge consumption of fuel and electricity. It can lead to the production of a high quantity of greenhouse gas emissions [8].

2.2 Aquaponics

Aquaponics is a term of aquaculture and hydroponics shown in figure 3. It is a process in which plants are cultivated within aquaculture systems wherein the fish provide the necessary nutrients to facilitate the development of the plants, and the name speaks for itself [10]. In this system, fish and plants are cultured together. Fish breeding in inner ponds produces nutrient-rich waste, which can be fed the plants in the vertical farming system. The plants filter and clean the recycled wastewater to be used in the fish ponds as a return. In Germany, there was a first supermarket with the greenhouse roof, which uses an aquaponic system, where bio-waste from fish are used as fertilizers of plants and serve as bio-filter in cleaning the water and return it to the fish tank. REWE green agricultural store is not only a supermarket but also a production facility in the middle of the city. The rooftop farm operated by ECF Farm Systems, the partner of REWE, can grow 800,000 basil plants every year through aquaponics [11]. Since aquaponics uses waste products from the fish REWE grows on site as fertiliser pesticides are applied in this process. Combining retail with a basil farm and fish farm, this store is the only one in Europe with a rooftop greenhouse. Another widely known aquaponic urban garden is in Sweden where local tropical product manufacture, such as mangoes and bananas, is done. Technological advances favorable to the region drive food sustainability through greenhouse operations.

2.3 Aeroponics

Aeroponics is an indoor horticulture technique that maximizes nutrient absorption and oxygenation by suspending the roots of plants in a nutrient-rich mist illustrated in figure 4. This farming technique is highly efficient and can be used to grow plants at significantly higher rates and yields than the majority of farming techniques [12].

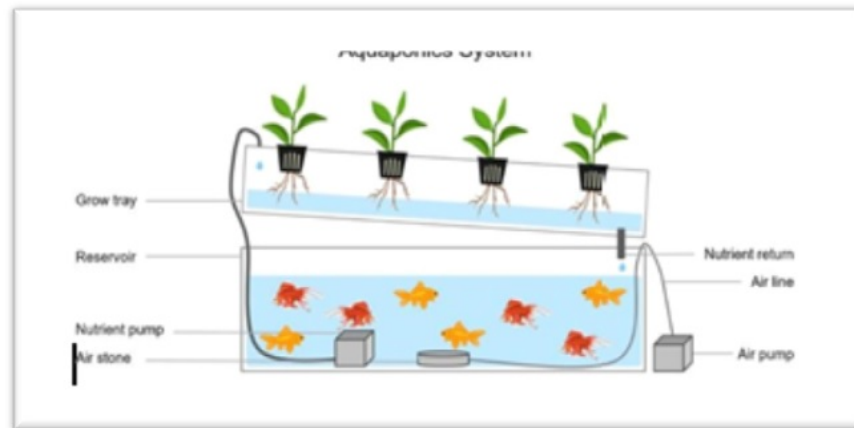


Figure 3: Aquaponic System

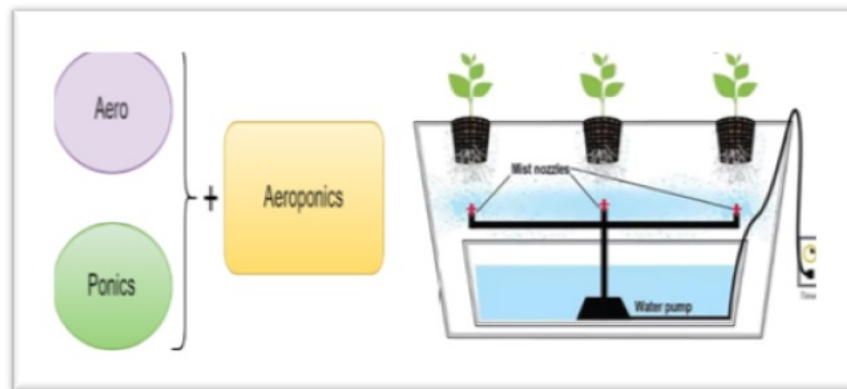


Figure 4: Aeroponic System

Aeroponics technology is novel, germane, and innovative. It can produce plants in colossal quantities and tree seedlings that have a connection with AM fungi and soil microorganisms in order to restore degraded land that is in the humid tropics. Aeroponically, plants are produced in an environment of thick vapour or air. A solution of water nutrient is sprayed onto the hanging roots of the plant. It is a high-pressure injector that has a micro inject nozzle with an electronic timer to perform the lower stems in a closed or semi-closed environment. It gives plants very oxygenated nutrients. However, misting cycles have to be adjusted to the plant species, since the roots dry very quickly due to exposing them to air and the skew of misting is quickly affected by the temperature outside which makes it less effective at pretty much low temperatures.

2.4 Comparative Study of Vertical Farming Techniques

Comparatively to conventional soil-based farming, this is the percentage of water conserved. While horizontal development calls for considerable area, space filling efficiency that is, the amount of space needed for crop development in vertical space is somewhat

different. The volume of energy needed to operate the system determines its energy efficacy. It evaluates the quantity of human labor required to operate and maintain the system independently. The inventory in the table also includes the requirements of crop varieties and resources finally. Thus, it is sensible to compare all three vertical farming techniques, since each system is different in terms of their growth ratio and specialty. Table 2 shows that crop species, environmental conditions, and system design all influence the yield and productivity of every of the above vertical approaches.

2.5 Optimizing Vertical Farming through Machine Learning and Internet of Things

Vertical farming is the ultimate expression of intelligent agriculture and marks a new horizon for precision agriculture, ushering in possibilities for sustainable agriculture that has been possible because of integration of machine learning with the technology of Internet of Things[14] and its models of vertical farming. Several formulas for machine learning and deep learning are applied in various vertical irrigation conditions. Moreover, artificial intelligence methods are there for predicting crop yield, tracking plant growth, and

Table 2: Comparative study of Other Vertical Approaches

Parameters	Hydroponic System	Aquaponic System	Aeroponic System
Energy Saving	High	High	High
Conserving Water Percentage	Up to 90%	Up to 90%	Up to 95%
Labor Work	High	High	Low
Productivity (output per square meter per year)	150-200 kg	50-80 kg	200-250 kg
Resource Requirements	Water rich in nutrients, lighting for growth, and a suitable growing medium (optional)	Water solution rich in nutrients, aquarium, planting bed, lighting for growth.	Misting nozzles deliver a nutrient-rich water solution for plant growth lights
Crop Varieties Grown	Many	Limited	More
Space Utilization Efficiency	Up to 90%	Up to 75%	Up to 95%

detecting disease as well as evaluating the quality of the samples. Along with these techniques, there is research over a huge volume of data obtained by IoT systems Advanced agricultural technologies are well-equipped to address future food supply challenges by fostering innovation within vertical farming systems[15]. The agricultural trends currently show that VF will introduce multiple dimensions of agricultural IoT devices. The outcome concerning the productivity and quality of the VF system will be better because of advanced technologies such as AI and ML[8].

3 Procedure for implementing Vertical Farming

The adoption of precision in indoor farming techniques, using automation powered by IoT and Machine Learning, needs to be a structured and transformative approach. The steps to be followed in the implementation process are as follows Steps-**•Define Objectives: Create particular goals for energy efficiency and yield improvement.****•Deploy Vertical Farming Infrastructure:** Install the necessary vertical towers or racks for crop development. **•Install Low-Energy LED Systems:** Build LED light systems designed for plant growth with energy savings. **•Installation of IoT Sensors:** Install sensors tracking all environmental factors including temperature, humidity, and light. **•Configuration of Cloud Data Transmission:** Set IoT sensor data forwarding to a centralized cloud-based system for real-time monitoring. **•Preprocess Data:** Normalize and clean data gathered from sensors for use in analysis. **•Feature Engineering:** Create features engineering using an energy-efficient and yield-oriented methodology to improve the system performance. **•Development of ML Models:** Use machine learning models that optimize farming practices. **•Energy Optimization:** Apply optimization techniques to improve energy usage in lighting, HVAC, and other critical systems **Problem Type & Yield Prediction:** It is the selection of a problem type, for instance, energy efficiency, or yield prediction. **Crop Growth Prediction:** Use predictive analytics to predict the growth of crops based on historical data and real-time data. **Real-Time Insights Monitoring:** Continuously track the performance of the system for adjustments.

Automate Controls: Implement automated systems for lighting, climate control, etc. **Alerts:** Generate alerts regarding anomalies or deviations in performance.

3.1 Iterate and Improve Models: Refine machine learning models based on feedback and outcomes.

The Internet of Things (IoT) is a term that is used in the agricultural industry to describe the utilization of sensors and other devices to transform every aspect and activity in agriculture into data. The Internet of Things (IoT) is one of the reasons why agriculture can supply a substantial amount of vital information. It also acts as a model for developments in agricultural 4.0 research and development. technological advancements and innovations associated with the Internet of Things are expected to greatly impact the agricultural sector, which will have extensive applications in the field of agriculture.

4 Conclusion

Ultimately, vertical farming has evolved into a workable answer to problems including limited land use, water scarcity, and climate change that beset conventional farming. Together with developments in agricultural systems, this study paper looks at the current technologies connected to vertical farming and their resource requirements. Moreover, a comparison of several vertical farming approaches shows that these approaches greatly improve the production of the most demanded food crops in a better way. Vertical farming has social and financial benefits in accordance with world food security as well as environmental ones above horizontal rice fields. Based on past research, several uses for IoT in vertical farms have been proposed for managing several utilities. We investigate in particular how artificial intelligence algorithms could be used in vertical farming. By means of smart and exact inside vertical farming, this study reveals fresh research prospects to explore and analyse indoor food crop production, so encouraging the creation of more sustainable and efficient food production systems. Every system has unique qualities and benefits, according to our study; so, the system of choice depends on numerous elements including the type of crop, the availability of space, and the investment cost.

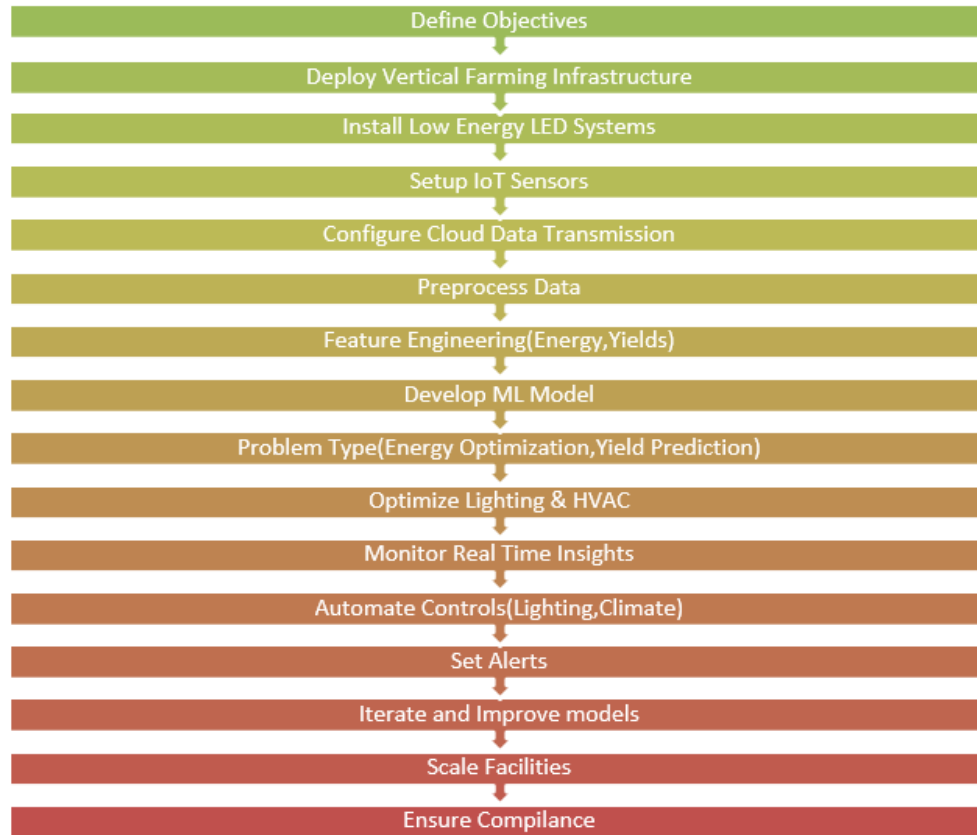


Figure 5: Implementing Procedure of Vertical Farming

Table 3: shows the accuracy of the algorithm for each vertical farming Technique.

VF Technique	Data Collection	ML Algorithm	Accuracy	Advantages	Applications
Hydroponics	IoT sensors (pH, EC, humidity, light, temperature)	SVR, ANN, RF	89.10%	Exact management of nutrient supply and environmental factors diminishes water consumption, boosts crop output, and reduces chemicals usage	Leafy greens, herbs, strawberries, tomatoes, cucumbers, peppers
Aquaponics	IoT sensors (pH, EC, light, humidity, temperature)	Decision Trees, RF, LR	91.28%	Minimizes water usage, utilizes fish waste as a nutrient source, high crop yield	Garlic, chives, carrots, mint, watercress
Aeroponics	IoT sensors (pH, EC, light)	XGBoost, RF	94.37%	Precise control over nutrient delivery	carrots, Herbs, cabbage, lettuce, tomatoes, leafy vegetables

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